

School of Engineering & Technology - SNPSU

INTERNET OF THINGS

MODULE – I

INTRODUCTION TO IoT & IoT ENABLING TECHNOLOGIES

QUESTION BANK

S. No	Question	CO	BTL	Marks
1 (a)	Define IoT and describe its key characteristics.	CO1	L1 (Remember) + L2 (Understand)	10M
1 (b)	Explain the role of “things” in IoT with examples.	CO1	L2 (Understand)	5M
2 (a)	List and briefly explain the various applications of IoT.	CO1	L1 (Remember) + L2 (Understand)	5M
2 (b)	Illustrate the layered architecture of IoT, highlighting its main components.	CO1	L2 (Understand) + L3 (Apply)	10M
3	Describe various physical devices used in IoT with neat diagrams.	CO1	L2 (Understand)	10M
4	Explain the working of the MQTT protocol, including its message format, control fields, and applications.	CO1	L2 (Understand)	10M
5	Illustrate the architecture of the CoAP protocol with its message format and applications.	CO1	L2 (Understand) + L3 (Apply)	10M
6	Discuss the working and applications of the HTTP protocol in IoT.	CO1	L2 (Understand)	10M
7	Explain the features, working principles, and applications of the AMQP protocol.	CO1	L2 (Understand)	10M
8	Illustrate the architecture, working, and applications of the XMPP protocol.	CO1	L2 (Understand)	10M
9 (a)	Describe the logical functional blocks of IoT with suitable examples.	CO1	L2 (Understand)	6M
9 (b)	Illustrate and explain the different communication models of IoT.	CO1	L2 (Understand)	10M
10	Differentiate between various communication APIs in IoT, supported with suitable diagrams.	CO1	L2 (Understand) + L4 (Analyze)	10M
11	Explain the architecture and applications of Wireless Sensor Networks (WSNs) with a diagram.	CO1	L2 (Understand)	8M
12	Illustrate the concept of cloud computing and compare different types of cloud services.	CO1	L2 (Understand) + L4 (Analyze)	10M
13	Explain the concept of Big Data analytics and analyze the different “Vs” with relevant examples.	CO1	L2 (Understand) + L4 (Analyze)	10M
14	Illustrate the various short-range communication protocols of IoT with examples.	CO1	L2 (Understand)	8M
15	Compare and explain various long-range communication protocols of IoT.	CO1	L2 (Understand) + L4 (Analyze)	6M
16	Explain the key components of Embedded Systems with block diagrams and applications.	CO1	L2 (Understand) + L3 (Apply)	10M